ADV: Calculus (Adv), C2 Differential Calculus (Adv)

L&E Differentiation (Y12)

Trig Differentiation (Y12)

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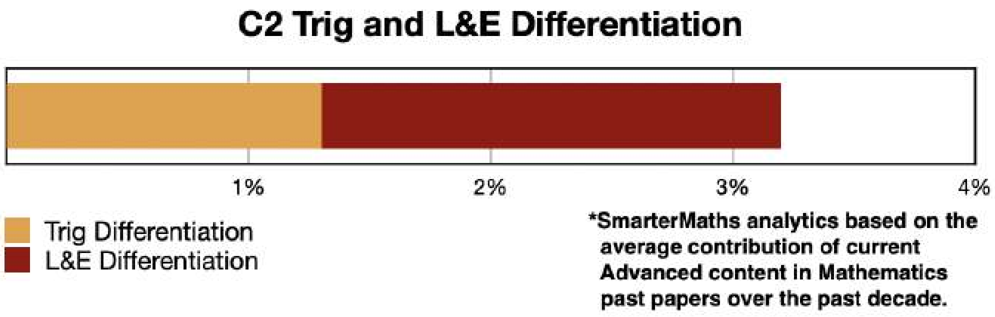
Exam Equivalent Time: 115.5 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*



HISTORICAL CONTRIBUTION

- *C2 Trig, Log and Exponential Differentiation* has contributed a healthy 3.2% to past Mathematics exams, on average over the past decade.
- This topic has been split into two sub-topics for analysis purposes: *1-Trig Differentiation (1.3%) and 2-Log and Exponential Differentiation (1.9%).*
- This analysis looks at *L&E Differentiation.*

HSC ANALYSIS - What to expect and common pitfalls

- This topic area is the definition of low lying fruit where well prepared students should expect to score highly.
- *L&E Differentiation* as a stand alone sub-topic is consistently examined and offers up some of the easiest marks in the Advanced exam, typically around the band 3 difficulty level.
- HSC Marker's Comments have noted that students who wrote out the quotient rule and all its inputs before substituting in values made less mistakes.
- The new syllabus explicitly mentions the differentiation of a^x and $\log_a x$ which are both covered in the database within SM-Bank questions.

Questions

1. Calculus, 2ADV C2 2016 HSC 5 MC

   What is the derivative of $\ln(\cos x)$?

   (A) $-\sec x$

   (B) $-\tan x$

   (C) $\sec x$

   (D) $\tan x$

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2. Calculus, 2ADV C2 2017 HSC 3 MC

   What is the derivative of e^{x^2}?

   (A) $x^2 e^{x^2-1}$

   (B) $2xe^{2x}$

   (C) $2xe^{x^2}$

   (D) $2e^{x^2}$

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3. Calculus, 2ADV C4 2018 HSC 5 MC

   What is the derivative of $\sin(\ln x)$, where $x > 0$?

   (A) $\cos\left(\dfrac{1}{x}\right)$

   (B) $\cos(\ln x)$

   (C) $\cos\left(\dfrac{\ln x}{x}\right)$

   (D) $\dfrac{\cos(\ln x)}{x}$

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4. Calculus, 2ADV C2 EQ-Bank 4 MC

If $f(x)=\log_2\left(x^{2x}\right)$, which expression is equal to $f'(x)$?

(A) $\dfrac{2}{x^{2x}\ln 2}$

(B) $\dfrac{2}{\ln 2}+2\log_2 x$

(C) $\log_2 x+\dfrac{2}{\ln 2}$

(D) $\dfrac{2}{\ln 2}\times\log_2\left(x^{2x-1}\right)$

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5. Calculus, 2ADV C2 2013 HSC 4 MC

What is the derivative of $\dfrac{x}{\cos x}$?

(A) $\dfrac{\cos x+x\sin x}{\cos^2 x}$

(B) $\dfrac{\cos x-x\sin x}{\cos^2 x}$

(C) $\dfrac{x\sin x-\cos x}{\cos^2 x}$

(D) $\dfrac{-x\sin x-\cos x}{\cos^2 x}$

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6. Calculus, 2ADV C2 2019 HSC 11b

Differentiate $x^2\sin x$. *(2 marks)*

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7. Calculus, 2ADV C2 2008 HSC 2aii

Differentiate with respect to x:

$x^2\log_e x$ *(2 marks)*

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8. Calculus, 2ADV C2 2012 HSC 12aii

Differentiate with respect to x.

$\dfrac{\cos x}{x^2}$. *(2 marks)*

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9. Calculus, 2ADV C2 2006 HSC 2aii

Differentiate $\dfrac{\sin x}{x+1}$ with respect to x. *(2 marks)*

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10. Calculus, 2ADV C2 SM-Bank 7

Let $f(x)=\dfrac{e^x}{(x^2-3)}$.

Find $f'(x)$. *(2 marks)*

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11. Calculus, 2ADV C2 SM-Bank 8

Let $y=(x+5)\log_e(x)$.

Find $\dfrac{dy}{dx}$ when $x=5$. *(2 marks)*

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12. Calculus, 2ADV C2 2019 MET1 1a

Let $y=\dfrac{2e^{2x}-1}{e^x}$.

Find $\dfrac{dy}{dx}$. *(2 marks)*

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13. Calculus, 2ADV C3 2005 HSC 2d

Find the equation of the tangent to $y=\log_e x$ at the point $(e,1)$. *(2 marks)*

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14. Calculus, 2ADV C2 2009 HSC 2aii

Differentiate with respect to x.

$(e^x+1)^2$. *(2 marks)*

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15. Calculus, 2ADV C2 2011 HSC 1d

Differentiate $\ln(5x+2)$ with respect to x. *(2 marks)*

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16. Calculus, 2ADV C2 2012 HSC 11d

Differentiate $\left(3+e^{2x}\right)^5$. *(2 marks)*

17. Calculus, 2ADV C2 2012 HSC 12ai

Differentiate with respect to x

$(x-1)\log_e x$ *(2 marks)*

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18. Calculus, 2ADV C2 2013 HSC 11d

Differentiate $x^2 e^x$ *(2 marks)*

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19. Calculus, 2ADV C2 2007 HSC 2ai

Differentiate with respect to x:

$\dfrac{2x}{e^x+1}$ *(2 marks)*

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20. Calculus, 2ADV C2 2015 HSC 11e

Differentiate $(e^x+x)^5$. *(2 marks)*

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21. Calculus, 2ADV C2 2015 HSC 11f

Differentiate $y=(x+4)\ln x$. *(2 marks)*

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22. Calculus, 2ADV C2 2017 HSC 11d

Differentiate $x^3 \ln x$. *(2 marks)*

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23. Calculus, 2ADV C2 2018 HSC 11g

Differentiate $\dfrac{e^x}{x+1}$. *(2 marks)*

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24. Calculus, 2ADV C2 2004 HSC 3aii

Differentiate with respect to x:

$(1+\sin x)^5$. *(2 marks)*

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25. Calculus, 2ADV C2 2006 HSC 2ai

Differentiate $x\tan x$ with respect to x. *(2 marks)*

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26. Calculus, 2ADV C2 2007 HSC 2aii

Differentiate with respect to x:

$(1+\tan x)^{10}$. *(2 marks)*

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27. Calculus, 2ADV C2 2009 HSC 2ai

Differentiate with respect to x:

$x\sin x$ *(2 marks)*

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28. Calculus, 2ADV C2 2010 HSC 1e

Differentiate $x^2\tan x$ with respect to x. *(2 marks)*

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29. Calculus, 2ADV C2 2010 HSC 2a

Differentiate $\dfrac{\cos x}{x}$ with respect to x. *(2 marks)*

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30. Calculus, 2ADV C2 2011 HSC 4a

Differentiate $\dfrac{x}{\sin x}$ with respect to x. *(2 marks)*

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31. Calculus, 2ADV C2 2013 HSC 11c

Differentiate $(\sin x-1)^8$. *(2 marks)*

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32. Calculus, 2ADV C2 2017 HSC 11c

Differentiate $\dfrac{\sin x}{x}$. *(2 marks)*

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33. Calculus, 2ADV C2 2008 HSC 2aiii

Differentiate with respect to x:

$\dfrac{\sin x}{x+4}$. *(2 marks)*

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34. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x:

$e^{\tan(2x)}$ *(2 marks)*

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35. Calculus, 2ADV C2 EQ-Bank 1

Differentiate $\log_2 x^2$ with respect to x. *(2 marks)*

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36. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x:

10^{5x^2-3x}. *(2 marks)*

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37. Calculus, 2ADV C2 EQ-Bank 3

Differentiate $3x\ 6^x$. *(2 marks)*

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38. Calculus, 2ADV C2 2019 MET1 1b

Let $f(x) = x^2 \cos(3x)$.

Find $f'\left(\dfrac{\pi}{3}\right)$. *(2 marks)*

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39. Calculus, 2ADV C2 SM-Bank 4

Differentiate $5^{x^2}\,5x$. *(2 marks)*

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40. Calculus, 2ADV C2 SM-Bank 6

Differentiate with respect to x:

$\log_e x^x$. *(2 marks)*

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41. Calculus, 2ADV C2 EQ-Bank 10

The function $f(\theta) = \sin^3(2\theta)$.

If $f'(\theta) = 6\cos(2\theta) - 6\cos^n(2\theta)$, find the value of n. *(2 marks)*

Worked Solutions

1. Calculus, 2ADV C2 2016 HSC 5 MC

$$\begin{aligned} y &= \ln(\cos x) \\ \frac{dy}{dx} &= \frac{-\sin x}{\cos x} \\ &= -\tan x \end{aligned}$$

$\Rightarrow B$

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2. Calculus, 2ADV C2 2017 HSC 3 MC

$$\begin{aligned} y &= e^{x^2} \\ \frac{dy}{dx} &= 2x\, e^{x^2} \end{aligned}$$

$\Rightarrow C$

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3. Calculus, 2ADV C4 2018 HSC 5 MC

$$\begin{aligned} y &= \sin(\ln x) \\ \frac{dy}{dx} &= \cos(\ln x) \times \frac{d}{dx}(\ln x) \\ &= \cos(\ln x) \times \frac{1}{x} \\ &= \frac{\cos(\ln x)}{x} \end{aligned}$$

$\Rightarrow D$

Worked Solutions

4. Calculus, 2ADV C2 EQ-Bank 4 MC

$$\begin{aligned} f(x) &= \log_2\left(x^{2x}\right) \\ &= 2x\log_2 x \\ &= \frac{2x\ln x}{\ln 2} \end{aligned}$$

$$\begin{aligned} f'(x) &= \frac{1}{\ln 2}\left(2x\cdot\frac{1}{x}+2\ln x\right) \\ &= \frac{2}{\ln 2}+\frac{2\ln x}{\ln 2} \\ &= \frac{2}{\ln 2}+2\log_2 x \end{aligned}$$

$\Rightarrow\ B$

5. Calculus, 2ADV C2 2013 HSC 4 MC

$$y=\frac{x}{\cos x}$$

Let $u=x \qquad u'=1$

$\quad v=\cos x \qquad v'=-\sin x$

$$\begin{aligned} \therefore\ \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} \\ &= \frac{\cos x\,1 - x(-\sin x)}{(\cos x)^2} \\ &= \frac{\cos x + x\sin x}{\cos^2 x} \end{aligned}$$

$\Rightarrow\ A$

6. Calculus, 2ADV C2 2019 HSC 11b

Using the product rule:

$$\frac{d}{dx}\left(x^2\sin x\right) = x^2\cdot\cos x + 2x\sin x$$

7. Calculus, 2ADV C2 2008 HSC 2aii

$$\begin{aligned} y &= x^2\log_e x \\ \frac{dy}{dx} &= x^2\cdot\frac{1}{x}+2x\cdot\log_e x \\ &= x+2x\log_e x \end{aligned}$$

8. Calculus, 2ADV C2 2012 HSC 12aii

$$y=\frac{\cos x}{x^2}$$

$u=\cos x \qquad u'=-\sin x$

$v=x^2 \qquad v'=2x$

Using the quotient rule,

$$\begin{aligned} \frac{dy}{dx} &= \frac{u'v - v'u}{v^2} \\ &= \frac{-\sin x\cdot x^2 - 2x\cdot\cos x}{x^4} \\ &= \frac{-x\sin x - 2\cos x}{x^3} \end{aligned}$$

9. Calculus, 2ADV C2 2006 HSC 2aii

$$y=\frac{\sin x}{x+1}$$

Using $\dfrac{d}{dx}\left(\dfrac{u}{v}\right)=\dfrac{u'v-uv'}{v^2}$

$u=\sin x \qquad v=x+1$

$u'=\cos x \qquad v'=1$

$$\therefore\ \frac{dy}{dx}=\frac{\cos x(x+1)-\sin x}{(x+1)^2}$$

10. Calculus, 2ADV C2 SM-Bank 7

Let $u = e^x \quad \Rightarrow \quad u' = e^x$

$v = \left(x^2 - 3\right) \quad \Rightarrow \quad v' = 2x$

$$f'(x) = \frac{e^x\left(x^2-3\right) - 2xe^x}{\left(x^2-3\right)^2}$$

$$= \frac{e^x\left(x^2-2x-3\right)}{\left(x^2-3\right)^2}$$

11. Calculus, 2ADV C2 SM-Bank 8

$$\frac{dy}{dx} = 1 \times \log_e x + (x+5) \cdot \frac{1}{x}$$

$$= \log_e x + \frac{x+5}{x}$$

$$\therefore \frac{dy}{dx} \mid_{x=5} = \log_e 5 + 2$$

12. Calculus, 2ADV C2 2019 MET1 1a

Method 1

$$y = 2e^x - e^{-x}$$

$$\frac{dy}{dx} = 2e^x + e^{-x}$$

Method 2

$$\frac{dy}{dx} = \frac{4e^{2x} \cdot e^x - \left(2e^{2x}-1\right)e^x}{\left(e^x\right)^2}$$

$$= \frac{4e^{3x} - 2e^{3x} + e^x}{e^{2x}}$$

$$= \frac{2e^{2x}+1}{e^x}$$

13. Calculus, 2ADV C3 2005 HSC 2d

$$y = \log_e x$$

$$\frac{dy}{dx} = \frac{1}{x}$$

At $(e, 1)$,

$$m = \frac{1}{e}$$

Equation of tangent, $m = \dfrac{1}{e}$, through $(e, 1)$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{e}(x - e)$$

$$y - 1 = \frac{x}{e} - 1$$

$$y = \frac{x}{e}$$

14. Calculus, 2ADV C2 2009 HSC 2aii

$$y = (e^x + 1)^2$$

$$\frac{dy}{dx} = 2(e^x+1)^1 \times \frac{d}{dx}(e^x+1)$$

$$= 2e^x(e^x+1)$$

15. Calculus, 2ADV C2 2011 HSC 1d

$$y = \ln(5x+2)$$

$$\frac{dy}{dx} = \frac{5}{5x+2}$$

16. Calculus, 2ADV C2 2012 HSC 11d

$$y = \left(3 + e^{2x}\right)^5$$

$$\begin{aligned}\frac{dy}{dx} &= 5\left(3 + e^{2x}\right)^4 \times \frac{d}{dx}\left(3 + e^{2x}\right)\\ &= 5\left(3 + e^{2x}\right)^4 \times 2e^{2x}\\ &= 10e^{2x}\left(3 + e^{2x}\right)^4\end{aligned}$$

17. Calculus, 2ADV C2 2012 HSC 12ai

$$\begin{aligned}y &= (x-1)\log_e x\\ \frac{dy}{dx} &= 1(\log_e x) + (x-1)\frac{1}{x}\\ &= \log_e x + 1 - \frac{1}{x}\end{aligned}$$

18. Calculus, 2ADV C2 2013 HSC 11d

Using the product rule

Let $u = x^2, \quad u' = 2x$

Let $v = e^x, \quad v' = e^x$

$$\begin{aligned}\frac{d(uv)}{dx} &= u'v + v'u\\ &= 2xe^x + x^2e^x\\ &= xe^x(x+2)\end{aligned}$$

19. Calculus, 2ADV C2 2007 HSC 2ai

$$y = \frac{2x}{e^x + 1}$$

$$u = 2x \qquad v = e^x + 1$$

$$u' = 2 \qquad v' = e^x$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2}\\ &= \frac{2(e^x+1) - 2x(e^x)}{(e^x+1)^2}\\ &= \frac{2e^x + 2 - 2x\cdot e^x}{(e^x+1)^2}\\ &= \frac{2(e^x + 1 - xe^x)}{(e^x+1)^2}\end{aligned}$$

20. Calculus, 2ADV C2 2015 HSC 11e

$$\begin{aligned}y &= (e^x + x)^5\\ \frac{dy}{dx} &= 5(e^x+x)^4 \times \frac{d}{dx}(e^x + x)\\ &= 5(e^x+x)^4 \times (e^x+1)\\ &= 5(e^x+1)(e^x+x)^4\end{aligned}$$

21. Calculus, 2ADV C2 2015 HSC 11f

$y = (x+4)\ln x$

Using the product rule

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x+4)\cdot\ln x + (x+4)\frac{d}{dx}\ln x\\ &= \ln x + (x+4)\frac{1}{x}\\ &= \ln x + \frac{4}{x} + 1\end{aligned}$$

22. Calculus, 2ADV C2 2017 HSC 11d

$y = x^3 \ln x$

Using the product rule:

$$\frac{dy}{dx} = 3x^2 \cdot \ln x + x^3 \cdot \frac{1}{x}$$
$$= x^2(3\ln x + 1)$$

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23. Calculus, 2ADV C2 2018 HSC 11g

$$y = \frac{e^x}{x+1}$$

Differentiate using quotient rule:

$$u = e^x \qquad v = x + 1$$
$$u' = e^x \qquad v' = 1$$

$$\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$$
$$= \frac{e^x(x+1) - e^x \cdot 1}{(x+1)^2}$$
$$= \frac{xe^x}{(x+1)^2}$$

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24. Calculus, 2ADV C2 2004 HSC 3aii

$$y = (1 + \sin x)^5$$
$$\frac{dy}{dx} = 5(1 + \sin x)^4 \times \frac{d}{dx}(\sin x)$$
$$= 5\cos x(1 + \sin x)^4$$

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25. Calculus, 2ADV C2 2006 HSC 2ai

i. $y = x \tan x$

Using product rule

$$\frac{d}{dx}(uv) = u'v + uv'$$
$$\therefore \frac{dy}{dx} = \tan x + x \times \sec^2 x$$
$$= x\sec^2 x + \tan x$$

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26. Calculus, 2ADV C2 2007 HSC 2aii

$$y = (1 + \tan x)^{10}$$
$$\frac{dy}{dx} = 10(1 + \tan x)^9 \times \frac{d}{dx}(\tan x)$$
$$= 10\sec^2 x\,(1 + \tan x)^9$$

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27. Calculus, 2ADV C2 2009 HSC 2ai

$$y = x\sin x$$
$$\frac{dy}{dx} = x\cos x + \sin x \times 1$$
$$= x\cos x + \sin x$$

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28. Calculus, 2ADV C2 2010 HSC 1e

$y = x^2 \tan x$

Using product rule:

$$\frac{d}{dx}(uv) = u'v + uv'$$
$$\frac{dy}{dx} = 2x\tan x + x^2\sec^2 x$$

COMMENT: There is no necessity to take out the common factor x in this case.

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29. Calculus, 2ADV C2 2010 HSC 2a

$$y = \frac{\cos x}{x}$$

Let $u = \cos x \qquad u' = -\sin x$

$\quad\;\; v = x \qquad\quad v' = 1$

Using quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2}\\ &= \frac{-\sin x \cdot x - \cos x \cdot 1}{x^2}\\ &= \frac{-x\sin x - \cos x}{x^2}\end{aligned}$$

MARKER'S COMMENT: A significant number of students did not know the quotient rule and many found assistance by writing out u, u', v, v' before going into calculations.

30. Calculus, 2ADV C2 2011 HSC 4a

$$y = \frac{x}{\sin x}$$

$u = x \qquad u' = 1$

$v = \sin x \qquad v' = \cos x$

Using $\dfrac{d}{dx}(uv) = \dfrac{u'v - uv'}{v^2}$,

$$\begin{aligned}\frac{dy}{dx} &= \frac{1 \cdot \sin x - x \cdot \cos x}{(\sin x)^2}\\ &= \frac{\sin x - x\cos x}{\sin^2 x}\end{aligned}$$

31. Calculus, 2ADV C2 2013 HSC 11c

$$\begin{aligned}y &= (\sin x - 1)^8\\ \frac{dy}{dx} &= 8(\sin x - 1)^7 \times \frac{d}{dx}(\sin x - 1)\\ &= 8(\sin x - 1)^7 \times \cos x\\ &= 8\cos x(\sin x - 1)^7\end{aligned}$$

32. Calculus, 2ADV C2 2017 HSC 11c

$$y = \frac{\sin x}{x}$$

Let $u = \sin x \qquad u' = \cos x$

$\quad\;\; v = x \qquad\quad v' = 1$

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2}\\ &= \frac{x\cos x - \sin x}{x^2}\end{aligned}$$

33. Calculus, 2ADV C2 2008 HSC 2aiii

$$y = \frac{\sin x}{x+4}$$

$u = \sin x \qquad u' = \cos x$

$v = x + 4 \qquad v' = 1$

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2}\\ &= \frac{\cos x(x+4) - \sin x}{(x+4)^2}\end{aligned}$$

34. Calculus, 2ADV C2 EQ-Bank 2

$$\begin{aligned}y &= e^{\tan(2x)}\\ \frac{dy}{dx} &= \frac{d}{dx}\tan(2x) \times e^{\tan(2x)}\\ &= 2\sec^2(2x) \cdot e^{\tan(2x)}\end{aligned}$$

35. Calculus, 2ADV C2 EQ-Bank 1

$$\begin{aligned} y &= \log_2 x^2 \\ \frac{dy}{dx} &= \frac{d}{dx}\left(\frac{\ln x^2}{\ln 2}\right) \\ &= \frac{1}{\ln 2}\cdot\frac{d}{dx}(\ln x^2) \\ &= \frac{1}{\ln 2}\cdot\frac{2x}{x^2} \\ &= \frac{2}{x\ln 2} \end{aligned}$$

TIP: The new Advanced reference sheet can be used here!

36. Calculus, 2ADV C2 EQ-Bank 2

$$\begin{aligned} y &= 10^{5x^2-3x} \\ \frac{dy}{dx} &= \ln 10\,(10x-3)\cdot 10^{5x^2-3x} \end{aligned}$$

TIP: The new Advanced reference sheet can be used here!

37. Calculus, 2ADV C2 EQ-Bank 3

$$\begin{aligned} y &= 3x\cdot 6^x \\ \frac{dy}{dx} &= 3\cdot 6^x + \ln 6\cdot 6^x\cdot 3x \\ &= 3\cdot 6^x(1+x\ln 6) \end{aligned}$$

COMMENT: See HSC exam reference sheet when differentiating 6^x.

38. Calculus, 2ADV C2 2019 MET1 1b

$$\begin{aligned} f(x) &= x^2\cos 3x \\ f'(x) &= x^2\cdot 3(-\sin 3x) + 2x\cos 3x \\ f'\left(\frac{\pi}{3}\right) &= \left(\frac{\pi}{3}\right)^2\cdot 3(-\sin\pi) + 2\left(\frac{\pi}{3}\right)\cos\pi \\ &= -\frac{2\pi}{3} \end{aligned}$$

39. Calculus, 2ADV C2 SM-Bank 4

$$\begin{aligned} y &= 5^{x^2}\cdot 5x \\ \frac{dy}{dx} &= \ln 5\cdot 2x\cdot 5^{x^2}\cdot 5x + 5^{x^2}\cdot 5 \\ &= 5^{x^2}(\ln 5\cdot 10x^2+5) \\ &= 5^{x^2+1}(\ln 5\cdot 2x^2+1) \end{aligned}$$

COMMENT: See HSC exam reference sheet when differentiating 5^x.

40. Calculus, 2ADV C2 SM-Bank 6

$$\begin{aligned} y &= \log_e x^x \\ &= x\log_e x \\ \frac{dy}{dx} &= x\cdot\frac{1}{x} + \log_e x \\ &= 1+\log_e x \end{aligned}$$

41. Calculus, 2ADV C2 EQ-Bank 10

$$\begin{aligned} f(\theta) &= \sin^3(2\theta) \\ &= (\sin(2\theta))^3 \end{aligned}$$

$$\begin{aligned} f'(\theta) &= 3\times 2\cos(2\theta)\times\sin^2(2\theta) \\ &= 6\cos(2\theta)\left(1-\cos^2(2\theta)\right) \\ &= 6\cos(2\theta) - 6\cos^3(2\theta) \end{aligned}$$

$$\therefore\ n = 3$$